

GURVIR SINGH

Third-Year Undergraduate Student

Computer Science & Engineering
(Artificial Intelligence & Data Science)
GNA University

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RESEARCH INTERESTS

- Machine learning and data science: data processing, analysis, and visualization
- Physics-informed machine learning and scientific computing (currently learning the basics)
- Intelligent systems: applying AI/ML methods within practical software systems

EDUCATION

GNA University

Expected 2028

B.Tech, Computer Science & Engineering (Artificial Intelligence & Data Science)

Third year

CGPA: 8.13/10

Current coursework: Data Visualization, Theory of Computation,
Software Engineering, Computer Graphics

SELECTED PROJECTS

AI Log Intelligence Platform (ongoing)

Python, Streamlit, Pandas, Scikit-learn, Plotly

- Streamlit application that detects and parses Linux auth/syslog and Apache access/error logs, with summary statistics, keyword search, Plotly charts, and an optional Gemini incident report.
- Label-encoded process, module, and host fields plus PID, message length, and hour feed an Isolation Forest anomaly detector; K-Means then clusters the flagged entries.

Neural Forge (AI Blender Assistant)

Python, Blender Python API, LLM APIs

- Blender add-on that turns natural-language prompts into Blender Python scripts using Gemini, OpenAI, local Ollama models, or any OpenAI-compatible endpoint through one API client.
- Optionally adds scene context (objects, active selection) to prompts, runs generated code, and sends runtime errors back to the model for an automatic fix.

College Knowledge Base Crawler

Python, Requests, BeautifulSoup, lxml, SQLite

- Multi-threaded crawler (worker pool and task queue) with robots.txt compliance, a shared rate limit, URL normalization, and retries with exponential backoff that respect Retry-After.
- Extracts page content and metadata to JSON, downloads linked documents, and records crawl state in SQLite, producing a structured dataset for a future RAG pipeline (not yet built).

CodeEditor (2nd-year project)

Java, Swing, MySQL, JDBC

- Desktop code editor with multi-tab editing, token-based syntax highlighting for Java, Python, and HTML, line numbers, and find and replace.
- Stores recent files, a code-snippet library, and open-tab sessions in MySQL through JDBC data-access objects, restoring the previous session on startup.

TECHNICAL SKILLS

Programming

Python, C, C++, Java

Data and ML

Machine learning, data processing, data visualization

Libraries and tools

NumPy, Pandas, Scikit-learn, Plotly, Streamlit, Git, GitHub

CURRENT FOCUS

- Strengthening my machine learning fundamentals, to apply ML across different fields.
- Learning how known physics can be built into model training, e.g. through the loss function, and keeping written research notes as I study physics-informed machine learning.
- Early idea, not implemented: whether physics-based models combined with machine learning could help reason about hidden energy states in public Formula 1 telemetry.